中央大學 NATIONAL CENTRAL UNIVERSITY DEPARTMENT OF ELECTRICAL ENGINEERING 電機工程學系

A SA-Based Floorplan Framework for Fixed Outline Floorplanning: Maximizing Utilization and Minimizing Routing Length

Advisor: Yu-Guang Chen

Students: Hsuan-Yu Weng、 Tsai-Chen Li、 Chia-Chien Cheng、 Ching-Sheng Chang

Abstract-- This research aims to optimize the placement of soft modules to maximize spatial utilization and minimize wirelength in fixed-outline floorplanning. We use FM partitioning to divide the chip into four quadrants and construct a Quad-Btree for initial placement. Then, we apply Fast-SA to find the optimal solution, resulting in a superior HPWL compared to ICCAD's best case in one of our cases.

I. Introduction

Floorplanning is crucial for VLSI design, directly impacting chip performance. Previous research focused on fixed module placement, while our study, allowing for module shape and size alteration, adds complexity.

Constraints for soft modules:

- No overlapping with other modules
- No exceeding the chip boundary
- Can be transformed into polygons

Purposes:

- Maximizing spatial utilization
- Minimizing wirelength

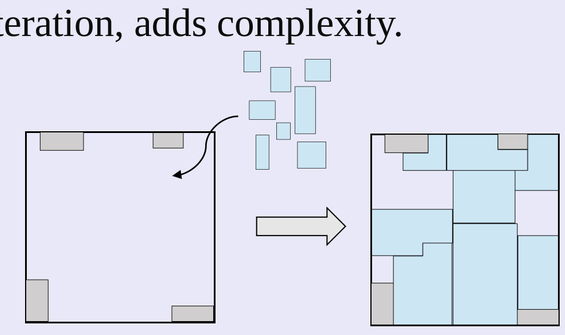


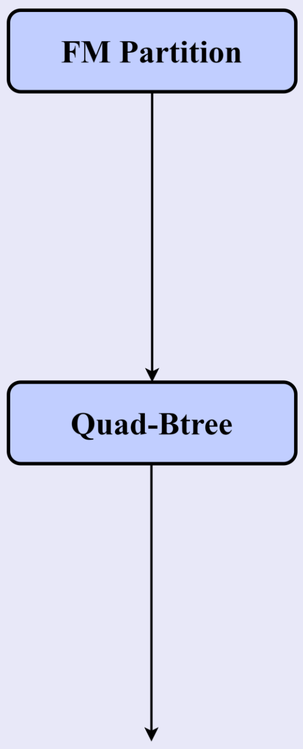
Fig. 1 Floorplanning

II. RESEARCH METHOD

We follow the process below to achieve the optimal placement results.

Fig. 2 Flow Chart of the Framework

A. Initialization



Allocate soft modules to suitable subregions to ensure connected modules are placed closely.

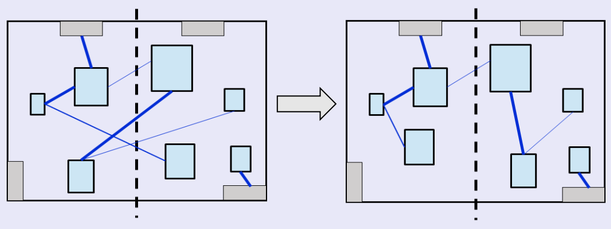


Fig. 3 Process of FM

Originate from the four corners of the floorplan and model the positions of modules.

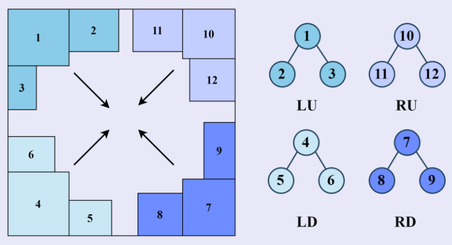


Fig. 4 Quad-Btree Diagram

B. Placement

Contour-based techniques are employed to avoid overlaps between soft modules and between soft modules and fixed modules.

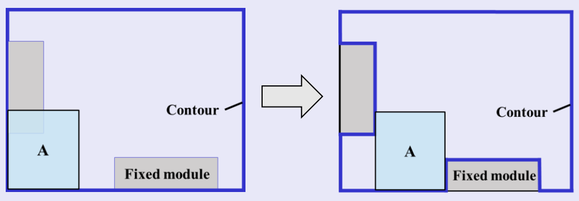


Fig. 5 Update Fixed Modules in Contour

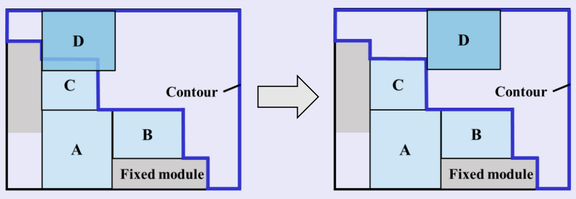
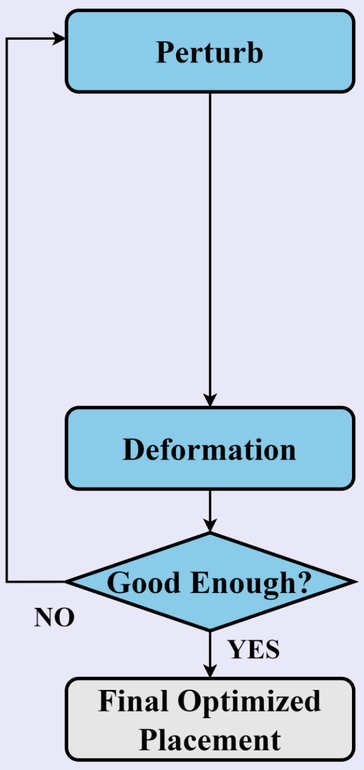


Fig. 6 Move x-Coordinate by Contour

C. Fast-SA Optimization



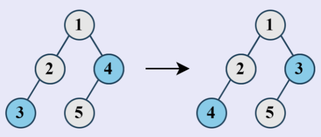


Fig. 7 Swap

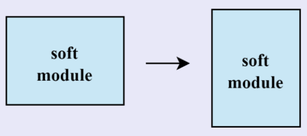


Fig. 9 Resize

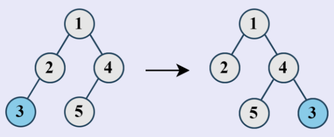


Fig. 8 Move

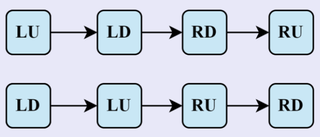


Fig. 10 Reorder

Cost function

$$F = \alpha \frac{W}{Wnorm} + (1-\alpha) P$$

W: current HPWL
P: the penalty factor for exceeding the boundary
Wnorm: the average of HPWL
α: user-defined parameter

III. RESULTS & DISCUSSION

We successfully legalized all cases through C++ programming within a runtime of 15 minutes, and utilized MATLAB for visualization.

Total HPWL

TABLE 1. the Results Compared to the Cases From ICCAD

Case	Rank 1	Rank 2	Rank 3	Rank 4	Rank 5	Ours	Our Rank
01	152,847,351.00	160,334,303.00	161,567,514.00	162,326,575.00	163,378,965.00	159,411,956.00	**1 - 2**
02	18,632,623.50	20,292,399.00	20,601,431.50	22,688,183.00	23,179,146.00	20,477,630.00	**2 - 3**
03	1,852,003.00	1,888,607.00	1,938,549.00	2,068,568.50	2,135,950.00	2,072,142.00	**4 - 5**
04	63,813,937.50	68,089,725.00	70,873,312.50	72,134,712.50	72,210,275.00	62,103,650.00	**<1**
05	16,058,950.00	16,774,600.00	17,966,700.00	18,632,550.00	19,247,350.00	19,129,000.00	**4 - 5**
06	35,741,550.00	37,895,150.00	40,173,950.00	42,011,500.00	42,644,800.00	38,627,450.00	**2 - 3**

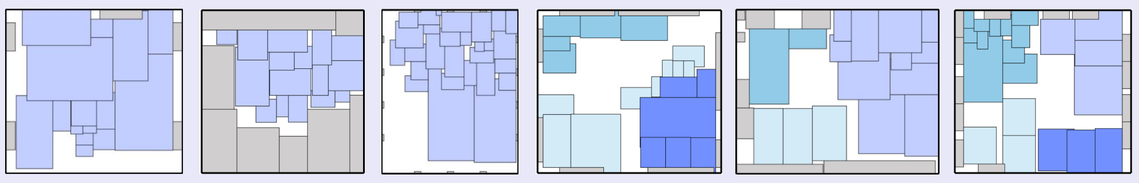


Fig.11 (a) Case01 (b) Case02 (c) Case03 (d) Case04 (e) Case05 (f) Case06

The Effectiveness of FM Partitioning

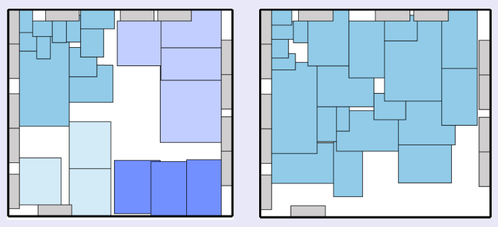


Fig. 12 Quad-Subregion vs. One-Subregion

TABLE 2. Classified Floorplanning Results

Case06	HPWL	HPWL Reduction
FM	38,627,450.00	-
From LD	48,017,100.00	24.31%
From LU	49,627,300.00	28.48%
From RU	46,342,850.00	19.97%
From RD	45,389,900.00	17.51%

The Effectiveness of Global Placement

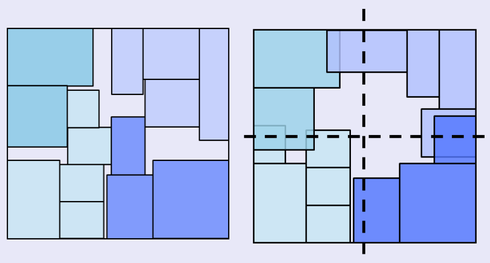


Fig. 13 Global Placement vs. Local Placement

TABLE 3. Partitioned Floorplanning Results

Case	Global Placement	Local Placement	HPWL Reduction
01	159,411,956.00	222,087,000.00	39.32%
02	20,477,630.00	25,051,300.00	22.33%
03	2,072,142.00	2,470,400.00	19.22%
04	62,103,650.00	79,185,700.00	27.51%
05	19,129,000.00	22,094,600.00	15.50%
06	38,627,450.00	47,455,100.00	22.85%

The Effectiveness of Fast-SA and Case-Based Parameter

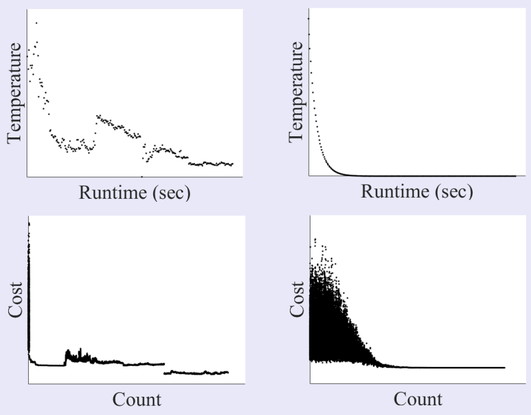


Fig. 14 Fast-SA vs. Classical-SA

TABLE 4. Fast-SA and Classical-SA Results

Case	Fast-SA	Classical-SA	HPWL Reduction
01	159,411,956.00	179,793,193.00	12.79%
02	20,477,630.00	50,411,942.50	146.18%
03	2,072,142.00	3,324,732.00	60.45%
04	62,103,650.00	66,340,112.50	6.82%
05	19,129,000.00	23,564,350.00	23.19%
06	38,627,450.00	47,986,100.00	24.23%

Given that varying parameters within the cost function influence the optimization outcome, we adapt them according to different cases.

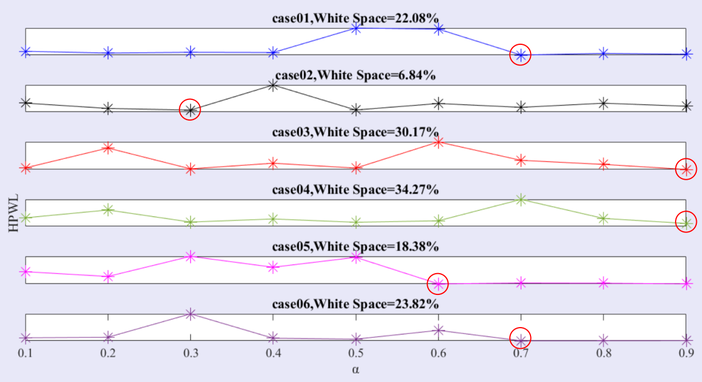


Fig. 15 HPWL Results Obtained with Different α

IV. CONCLUSION

We present an algorithm using our Quad-Btree and Fast-SA framework for fixed-outline floorplanning, ensuring no module overlaps. Our framework outperforms Classical B*-tree and Classical-SA, and adjusting the cost function's α parameter based on whitespace ratios further reduces wirelength. Ultimately, we surpass ICCAD's optimal solution in one case.

V. REFERENCE

1. Sheblaev, M. V., & Sheblaeva, A. S. (2018). A method of improving initial partition of Fiduccia–Mattheyses algorithm. Lobachevskii Journal of Mathematics, 39, 1270-1276.
2. Comer, D. (1979). Ubiquitous B-tree. ACM Computing Surveys (CSUR), 11(2), 121-137.
3. Tung-Chieh, C., & Yao-Wen, C. (2006). Modern floorplanning based on B*-tree and fast simulated annealing. IEEE Transactions on computer-aided design of integrated circuits and systems, 25(4), 637-650.

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